



PH 625: Data Analysis in Particle Physics

Course Code	PH 625
Course Name	Data Analysis in Particle Physics
L-T-P-C	2-0-4-4
Prerequisites	PH 621 (Computational Methods for Physicists), PH612 (Nuclear and Particle Physics)
Intended for	UG/PG/I-PhD/PhD elective
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

- **Data in Experiments, Particle Physics [4 Lectures]**
 - Brief overview of experiments in High Energy Physics. RHIC & CERN experiments.
 - Data from Experiments: Pulse processing, Timing and energy resolution, Tracking, Particle Identification (PTD).
 - Analysis Methods: Acceptance, Efficiency, Error Calculations, Observable quantities.
 - Data Structure/Type/Fonnat and algorithms from experiments and handling.
- **Recalls [10 Lectures]**
 - Brief Overview of c++: Program Organization and Control Structures loops, arrays, and finction, Error, Accuracy, and Stability. Transition from C to c++.
 - Brief Overview of numerical analysis in c++: Curve Fitting, ROOT finding, Integration and differentiation, Interpolation and Extrapolation
- **ROOT Framework and Familiarities [in root/pyroot/rootpy] [15 Lectures]**
 - Introduction to ROOT: Is an object-oriented programming framework based on c++ developed by CERN. Originally designed for particle physics, but it is used in other applications such as astronomy and data mining. ROOT has capability to work in any field and potential to scale globally.
 - ROOT installation: ROOT is available on Linux, Mac. and (as a beta release) on Windows. The latest stable ROOT re lease is updated.
 - ROOT preliminaries:
 - Mathematical foundation, input, output, functions
 - Histograms handling: Writing and reading: Basic, Binning, Statistical analysis: ID, 20,30
 - Tree handling: Writing and reading of the key feature of root
 - Libraries and useful tools
 - Fitting data: Formulas, Reading data, Writing data, TFI functions, Fittings.
- **Visualization in ROOT [in root/pyroot/rootpy] (10 Lectures)**
 - Histograms: 1D, 2D, 3D and asymmetric binning
 - Trees, TProfiles, TBrower
 - Graph Plotting, TCanvas, TGraph, TGraphError, Graphs with asymmetric error.
 - Markers and legends
 - Histograms fitting with functions
- **Statistical Analysis and Error Methods (7 Lectures)**

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- Statistical Analysis
 - Statistical Error: Gaussian Method, Delta Theorem, Bootstrap method
 - Systematic error Estimation
 - **Data Generation and Models** (5 Lectures)
 - Concept of Simulation
 - Random Generators
 - Monte Carlo Simulation and Data Generation
 - **Data Analysis: Class Project** (5 Lectures)
 - Astro Physics data analysis High Energy Physics data Analysis

Text Books

1. Yashavant Kanetkar, Let Us C: Authentic guide to C programming.
2. Yashavant Kanetkar, Let Us C++.
3. <https://IROOT.cern/manual/>
4. David J. Griffiths, Introduction to Elementary Particles
5. William R. Leo, Techniques for Nuclear and Particle Physics Experiments
6. Fred James, Statistical Methods in Experimental Physics

References

1. Yashavant Kanetkar, Let Us C++ Solutions
2. ROOT.cern.ch
3. cern.ch
4. rhic.bnl.gov